NAME: _____ Geometry (FL B.E.S.T.)

12.4 Permutations and Combinations

A **permutation** counts ordered arrangements — position matters, so swapping two chosen items creates a different outcome: ${}_nP_r = \dfrac{n!}{(n-r)!}$. A **combination** counts unordered selections — swapping two chosen items gives the *same* outcome (a committee is a committee no matter what order you name its members): ${}_nC_r = \dfrac{n!}{r!(n-r)!} = \dfrac{{}_nP_r}{r!}$. The single most important skill here is deciding *which one applies* before computing anything — ask "does order/role matter?" This worksheet also combines permutations/combinations with the Fundamental Counting Principle, restrictions (a specific person must/must not be included or placed), and probability.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Evaluate $4!$.

2. Find ${}_6P_3$.

3. Find ${}_8C_3$.

4. The finishing order of 5 of 12 runners in a race: permutation or combination?

5. Choosing 5 of 12 runners for a relay team (no order): permutation or combination?

6. Find ${}_4P_4$.

7. Find ${}_5C_1$.

8. Find ${}_9C_2$.

9. Find ${}_8P_2$.

10. Evaluate $0!$.

11. If ${}_nC_3 = 20$, find ${}_nP_3$.

12. In how many ways can 4 of 6 different novels be arranged on a shelf?

NAME: _____ Geometry (FL B.E.S.T.)

13. In how many ways can 4 of 6 people be chosen for an unranked group?

14. 6 people sit in a row. Two specific people, X and Y, must sit at the two ends (in either order). How many seatings are possible?

15. A lottery requires picking 5 correct numbers from 40 (order doesn't matter), where ${}_{40}C_5 = 658{,}008$. Find the probability of winning with one ticket.

✓ Answer Key

1. 24
2. 120
3. 56
4. Permutation
5. Combination
6. 24
7. 5
8. 36
9. 56
10. 1
11. 120
12. ${}_6P_4 = 360$
13. ${}_6C_4 = 15$
14. $2 \times 4! = 48$
15. $\frac{1}{658{,}008}$